Identity-Disjoint Iris Recognition Using Hybrid Quantum Metric Learning and Few-Shot Evaluation

Project Report [Phase-I]

Submitted in partial fulfillment for the award of the degree of

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Indian Institute of Information Technology

Bhopal (M.P.)



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Certificate

This is to certify that the work embodied in this report entitled "**Identity-Disjoint Iris Recognition Using Hybrid Quantum Metric Learning and Few-Shot Evaluation**" has been satisfactorily completed by **Aditya Chaurasia (Scholar No. 24U020143)** and **Priyanshu Sharma (Scholar No. 24U020144)**. It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal**, for the partial fulfillment of the Bachelor of Technology degree during the academic year 2026–27.

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Declaration

We hereby declare that the project report entitled “Identity-Disjoint Iris Recognition Using Hybrid Quantum Metric Learning and Few-Shot Evaluation”, presented in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is an authentic documentation of our original work carried out under the able guidance of Dr. Meenakshi Choudhary. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted, in part or in whole, for the award of any degree or professional diploma in any other institute or organization.

We declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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Area of Work

This project is based on machine learning, deep learning and quantum machine learning for iris recognition. The goal of the project is to develop and test various models for identifying an individual by their iris pattern.

The project begins with dataset exploration and image preprocessing in order to prepare the iris images for model training. Various methods are then applied for feature extraction and to enhance inter-person discrimination.

We apply and study several methods, such as convolutional neural networks (CNNs), Siamese networks and quantum-based approaches. A hybrid quantum-classical approach is also studied to explore the benefits of combining classical deep learning with quantum algorithms.

Finally, the models are compared using suitable performance metrics and experiments involving limited training images. The overall aim is to understand the effectiveness, advantages and limitations of these approaches for developing reliable biometric recognition systems.

Table of Contents

S. No.	Title	Page No.
	Certificate	2
	Declaration	3
	Acknowledgement	4
	Area of Work	5
	Abstract	7
	List of Figures	8
	List of Tables	9
1.	Introduction	10
2.	Literature Review	12
3.	Problem Definition and Objectives	15
4.	Proposed Methodology and Work Description	17
5.	Proposed Algorithms	25
6.	Proposed Flowchart / Block Diagram	27
7.	Tools, Technologies and Implementation Details	29
8.	Results, Discussion and Comparative Analysis	32
9.	Conclusion and Future Scope	38
10.	References	39
11.	Plagiarism Report	41

Abstract

Reliable iris recognition needs a feature representation that can capture unique iris patterns while still working well when image quality or capture conditions change. It should also be able to recognize identities that were not seen during training. In this project, we developed and evaluated an identity-disjoint iris verification system using the CASIA-Iris-Thousand dataset. The work was carried out in a series of steps: dataset exploration, iris preprocessing and normalization, identity-disjoint data splitting, a ResNet34 classification baseline, Siamese network-based metric learning, dimensionality reduction from 128 to 8 dimensions, quantum circuit experiments, hybrid quantum-classical metric learning, few-shot evaluation and ablation studies.

During dataset exploration, we found that the dataset contains 20,000 near-infrared grayscale iris images from 1,000 subjects and 2,000 eye identities, with 10 images available for each eye identity. In the preprocessing stage, pupil and limbus boundaries were detected, the iris region was normalized into 64 × 256 polar strips and image enhancement was applied. All 20,000 images were processed successfully without any failures. For training and evaluation, we used an identity-disjoint split containing 14,000 training images, 3,000 validation images and 3,000 test images. These belonged to 1,400, 300 and 300 unique identities respectively, with no identity overlap between the splits.

The Siamese network with a 128-dimensional embedding achieved a test accuracy of 79.28%, an area under the ROC curve (AUC) of 0.8797 and an equal error rate (EER) of 21.92%. After reducing the representation to 8 dimensions using a classical projection method, the model achieved an accuracy of 78.58%, an AUC of 0.8747 and an EER of 22.18%. In the hybrid quantum-classical experiments, the standard hybrid model achieved an accuracy of 79.90% and an AUC of 0.8813, while the product variant achieved an accuracy of 79.42% and an AUC of 0.8803. A later ablation study used a larger shared pair set and identity-level bootstrap confidence intervals. Since its evaluation protocol was different, those results are reported separately rather than being combined directly with the earlier results.

The study also examines Rank-1 and Rank-5 identification performance under 1-shot, 3-shot and 5-shot enrollment settings, along with false acceptance and false rejection trade-offs and the uncertainty in model comparisons. The experimental results show that the proposed 8-dimensional hybrid representation obtained results comparable with classical methods and improved some verification metrics, but they do not support the claim that quantum methods outperform classical methods in all cases. The contributions of this work are the use of strict identity separation, reproducibility and careful evaluation under various experimental settings.

List of Figures

Figure	Description	Page No.
1	Dataset distribution: eye side and number of samples per eye identity	18
2	Sample near-infrared iris images from the dataset	18
3	Sampled image quality metric profiles	18
4	Iris segmentation, normalization and enhancement stages	19
5	Identity-disjoint training, validation and test split	20
6	Classical ResNet34 training and validation curves	20
7	Siamese metric-learning training curves	21
8	Classical 128-D to 8-D projection results	22
9	Quantum circuit design sweep results	22
10	Frozen-feature classical and quantum head comparison	23
11	End-to-end hybrid model training curves	23
12	Hybrid model test ROC curves (full and zoomed views)	24
13	Few-shot evaluation curves for 1-, 3- and 5-shot protocols	24
14	System block diagram	27
15	Frozen-feature classical and quantum head comparison	33
16	End-to-end hybrid model training curves	34
17	Hybrid model test ROC curves (full and zoomed views)	34
18	Verification ROC curves and EER confidence intervals	35
19	Paired ablation effects with confidence intervals	35
20	Few-shot evaluation curves for 1-, 3- and 5-shot protocols	36
21	Rank-1 performance by enrollment shots and FAR/FRR trade-off	36

List of Tables

Table	Description	Page No.
1	Dataset composition reported by the exploration notebooks	17
2	Identity-disjoint split used for the metric-learning experiments	19
3	Research workflow	29
4	Evaluation metrics and interpretation	30
5	Key experimental findings	32
6	Quantum circuit and frozen-feature control comparison	33
7	Final paired-pair verification results	35
8	Limitations and risk mitigation	37

1. Introduction

1.1 Background

Biometric recognition is used to identify or verify a person based on physical or behavioral characteristics [1]. Among different biometric traits, the iris is considered highly reliable because it contains unique texture patterns that differ for every individual [2], [3]. One of the other benefits of an iris recognition system is that it does not require any physical contact. However, iris recognition systems face many complications, including variations in lighting, focus, pupil size, occlusions, sensor variations and image quality [1]. Traditional iris recognition systems usually follow a series of steps: locating the pupil and iris boundaries, normalizing the iris region, extracting texture features and then comparing those features to determine whether two iris images belong to the same person [2], [3]. Biometric traits can also be combined to improve reliability [4], and the security and privacy of stored biometric templates remain important concerns [5].

With the rise of deep learning, iris recognition systems have become more data-driven, since deep learning models can learn the most useful features directly from the images. CNNs are the most popular deep learning architecture for image recognition tasks, as they are capable of learning both low-level and high-level visual features [6], [7]. Metric learning techniques are often combined with CNNs for recognition, as they learn an embedding space in which images of the same identity are close together while images of different identities are separated [8]–[10]. Siamese networks are well suited to verification tasks, as they learn directly from pairs of images and are not limited to a fixed set of identities during training [10]. However, high performance can be misleading if the same identities appear in both the training and testing data. To avoid this problem and make the evaluation more realistic, this project uses a strict identity-disjoint split, ensuring that identities in the test set are never seen during training.

1.2 Motivation

In this project, we investigate whether a small 8-dimensional iris representation can still retain sufficient information for verification, and whether a trainable quantum circuit can outperform existing machine learning heads [11], [12]. Several models of varying complexity were used and compared: a ResNet34 classifier, a 128-dimensional Siamese network, an 8-dimensional classical projection, a quantum circuit with frozen features and several end-to-end hybrid quantum-classical models.

The point of the project was not to show that quantum models always outperform classical models. The intention was to compare various techniques fairly and to analyze the costs and benefits of embedding size, model complexity, verification accuracy and few-shot recognition, since the performance of variational quantum models depends on factors such as data encoding [13]. All experiments were conducted under an identity-disjoint evaluation protocol to ensure a fair comparison between the different approaches.

1.3 Summary

The work in this project is organized into 10 notebooks. Notebooks 1 to 3 cover dataset exploration and quality checking, iris preprocessing, and identity-disjoint data splitting with few-shot protocol creation. Notebooks 4 to 10 cover model development and evaluation: the classical ResNet34 baseline, Siamese metric learning, dimensionality reduction, quantum circuit experiments, hybrid quantum-classical models, few-shot evaluation and ablation analysis with bootstrap statistics [14]. All results shown in this report are taken directly from the notebook outputs. Since some notebooks use different evaluation protocols and pair sets, their results are reported separately and are not compared directly.

1.4 Scope

This project is experimental and research-oriented. It includes image quality analysis, iris preprocessing and normalization, identity-disjoint data splitting, representation learning, verification testing, few-shot identification, quantum circuit experiments and statistical assessment, following standard biometric evaluation practice [1], [15]. The project is not meant to be a complete commercial biometric system. It does not cover real-time deployment, security certification, extensive user evaluation or performance on multiple datasets. All results and conclusions are based only on the dataset and evaluation protocols used in the project notebooks.

1.5 Contributions

The main contributions of this work, which builds on established iris recognition and metric learning practice [2], [8], are listed below.

- Auditing the CASIA-Iris-Thousand image set and its class balance before training the models.
- Recording iris segmentation and normalization into 64 × 256 grayscale strips, together with the success rate of the preprocessing.
- Building identity-disjoint training, validation and test partitions to measure generalization to new identities that are not in the training data.
- Comparing a ResNet34 classifier, Siamese metric learning and a compact classical projection.
- Designing and validating an 8-qubit variational circuit and comparing it with classical heads on matched embeddings.
- Training end-to-end hybrid variants and assessing verification, few-shot identification and paired ablations.

2. Literature Review

2.1 Iris Recognition and Normalization

Daugman's iris recognition method showed that accurate recognition depends on both good feature matching and proper iris preprocessing [2], [3]. To handle variations in pupil size and shape, the iris region is normalized to a fixed rectangular region. In our system, the preprocessing includes iris boundary detection, polar normalization and image enhancement, and a small 64 × 256 grayscale iris strip is used for further processing. Segmentation is a key step in this pipeline, and deep learning-based segmentation methods have also been widely studied [16].

The dataset has 20,000 iris images from 1,000 persons and 2,000 eye identities, with 10 images per eye identity [17]. The dataset is balanced in terms of samples per identity and can therefore be used to train and test all identities equally. The dataset exploration and quality check notebooks indicated that the images are readable and that no duplicate image paths were found [17].

2.2 CNN-Based Features

CNNs learn features from input images through multiple layers, initially learning simple features and, over subsequent layers, more complex ones [6], [18], [19]. For this project, ResNet34 is used as the backbone, as its residual connections allow deep models to be trained effectively [7]. The network extracts features from the images that are used for the classification or verification task. A baseline based on ResNet34 was trained on a 1,400-class classification task with an ArcFace-style loss function [20]. ArcFace encourages separation between classes and results in more discriminative features, which we use as our baseline model. Unlike the later identity-disjoint verification experiments, this baseline uses the same set of identities for training and testing within its own data split and is therefore evaluated as a closed-set classification task [20].

2.3 Metric Learning and Pairwise Verification

Metric learning learns features so that images of the same identity are closer together while images of different identities are farther apart [8], [9]. Siamese networks are commonly used for this purpose because they compare image pairs and are well suited to verification tasks [10]. In this project, a ResNet34-based Siamese network was used to generate 128-dimensional iris embeddings. The model achieved an AUC of 0.8797 and an EER of 21.92% on the test set. To evaluate performance, metrics such as accuracy, AUC, EER, false acceptance rate (FAR) and false rejection rate (FRR) were used [15]. These metrics provide a more complete understanding of verification performance than accuracy alone.

2.4 Limited-Sample Learning and Subject-Independent Evaluation

In this project, few-shot learning refers to the recognition of new identities based on only a few training samples [21], [22]. Each new identity is enrolled with 1, 3 or 5 iris images and tested on query images.

To make a fair comparison, all identities were kept entirely separate across the training (1,400 identities), validation (300 identities) and test (300 identities) sets. The few-shot evaluation uses the same support and query sets for all models, which provides a fair comparison [21].

2.5 Low-Dimensional Feature Encoding

Compressing the output vector may not have a great impact on performance, but it can save storage and allow faster comparisons [8]. The 128-dimensional embedding was reduced to 8 dimensions, which is a substantially smaller size. The results indicated that only a small decrease in accuracy and AUC was observed, which showed that the 8-dimensional embedding can still retain most of the crucial iris information while giving a much smaller vector to work with.

2.6 Variations in Quantum Algorithms

Quantum machine learning merges quantum circuits with classical machine learning models [11]. In hybrid methods, the data is encoded into a quantum circuit, processed through trainable quantum gates and then measured in order to generate features or predictions [12], [23]. Since quantum models can be affected by factors such as data encoding and circuit design [13], [24], they should be compared with strong classical baselines rather than assumed to perform better.

This project uses an 8-qubit quantum circuit with an 8-dimensional input feature vector. The circuit takes the input features and applies trainable quantum operations to generate an 8-dimensional output feature vector [12]. Several tests were performed to ensure that the circuit and the training process were working correctly. Since all experiments were carried out on a quantum simulator running on a classical computer, the results show that the approach works, but they do not demonstrate an advantage over classical methods or on real quantum hardware [11].

2.7 Statistical Comparison and Limitations of Point Estimates

Small differences between models may arise from random variation in the test data. Therefore, it is important to measure not only performance scores but also the uncertainty in those scores [14]. In this project, the final ablation study used 1,000 bootstrap samples at the identity level and compared models on the same test pairs to ensure a fair comparison [14].

The results show that there is no single best model for every metric. For example, one model may have a lower EER while another has a lower FAR, and a compact model may sacrifice a small amount of accuracy for a much smaller embedding size [15]. It is therefore important to interpret performance values together with their confidence intervals [14].

2.8 Research Gap and Study Contribution

In this project, we explore the performance of a small 8-dimensional iris embedding with various quantum and classical methods. The experiments include a classical linear head, the standard hybrid quantum-classical model and a quantum product variant. Other experiments include circuit design experiments and ablation experiments [12], [24].

The purpose is to allow an unbiased comparison of these approaches in as controlled a setting as possible [11]. The results give an impression of the trade-off between model complexity, embedding size and recognition accuracy. These results do not imply that quantum models outperform classical models, nor do they reveal any quantum advantage on real hardware.

3. Problem Definition and Objectives

3.1 Problem Definition

The aim of the project is to decide whether two iris images are from the same eye. The system is trained on image features and generates a score indicating whether the two images match. To test the system accurately, the identities in the test set are entirely different from those used in training.

The project further explores few-shot identification, where a new identity is enrolled from 1, 3 or 5 iris images. The core aim is to see whether the compact 8-dimensional hybrid quantum-classical representation can perform similarly to classical methods.

3.2 Objectives

1. Analyze the dataset for the number of images, classes, image quality and duplicates.
2. Perform preprocessing and normalization of the iris images.
3. Create identity-disjoint training, validation and test sets.
4. Build a ResNet34 baseline, test it and produce a baseline Siamese network.
5. Reduce the embedding size from 128 to 8 dimensions and analyze the impact on performance.
6. Develop and compare quantum and classical models using the same embeddings.
7. Evaluate the few-shot recognition performance of the different models.
8. Perform fair comparisons using shared test data and statistical analysis.

3.3 Research Questions

1. How much performance is lost when the embedding size is reduced from 128 dimensions to 8 dimensions?
2. Does the quantum model perform better than the classical model when both use the same input features?
3. Does the performance of the quantum and classical techniques improve when a complete end-to-end model is trained?
4. How accurately do the models perform when a new identity is introduced with only 1, 3 or 5 images?
5. Which performance differences remain meaningful after applying significance tests and confidence intervals?

3.4 Expected Outcomes

The outcome of the project relies on a fair and consistent comparison of classical and quantum-based models. The experiments should be carried out across distinct training and testing identities with a standard evaluation process and performance metrics. Results should be

presented using confidence intervals and appropriate statistical analysis. When two models are very close, no clear winner should be claimed.

4. Proposed Methodology and Work Description

4.1 Overall Pipeline

The project follows a step-by-step approach, which is summarized in Figure 14. First, the dataset is explored and a quality check is performed, followed by iris image preprocessing, including localization, normalization and enhancement. The data is then partitioned into identity-disjoint training, validation and test sets for the verification and few-shot experiments. After preprocessing, a ResNet34 baseline, a Siamese metric-learning model and a compact 8-D classical embedding were trained, followed by quantum circuit experiments and hybrid quantum-classical models. Finally, the models are evaluated using verification, few-shot identification and ablation studies, with statistical analysis used to compare their performance fairly.

4.2 Dataset Exploration and Quality Assessment

The dataset exploration and quality check confirmed that the dataset contains 20,000 iris images from 1,000 subjects and 2,000 eye identities, with 10 images per identity, as summarized in Table 1. No corrupted or duplicate images were found. The eye-side distribution and the number of samples per eye identity are shown in Figure 1, and sample near-infrared iris images are shown in Figure 2.

Table 1. Dataset composition reported by the exploration notebooks

Property	Notebook-reported value
Original images	20,000 near-infrared grayscale images
Subjects	1,000
Eye identities	2,000 (left and right eyes treated separately)
Images per subject	20
Images per eye identity	10
Original image dimensions	640 × 480 pixels
Eye-side distribution	10,000 left; 10,000 right
Image-path duplicates / unreadable originals	0 duplicates; 0 invalid in the separate quality audit

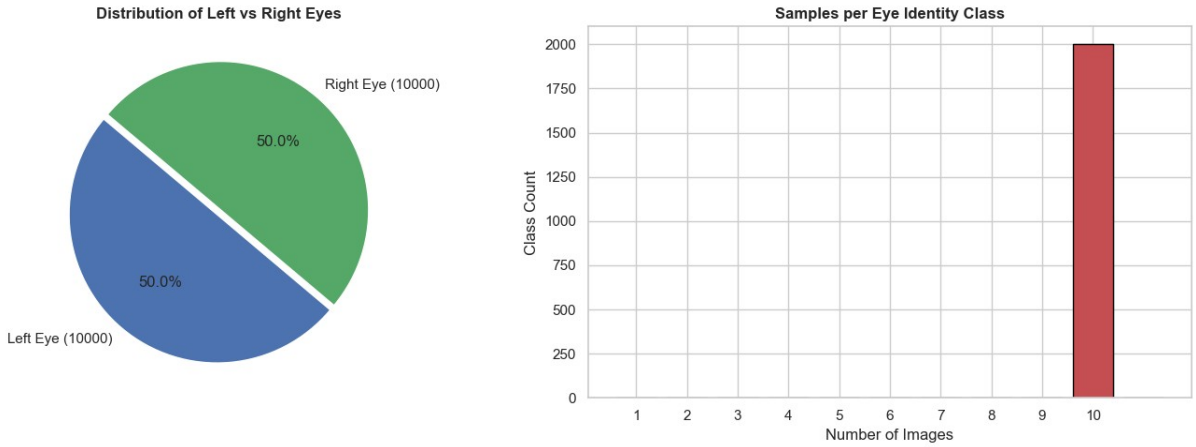


Figure 1. Dataset distribution: eye side and number of samples per eye identity

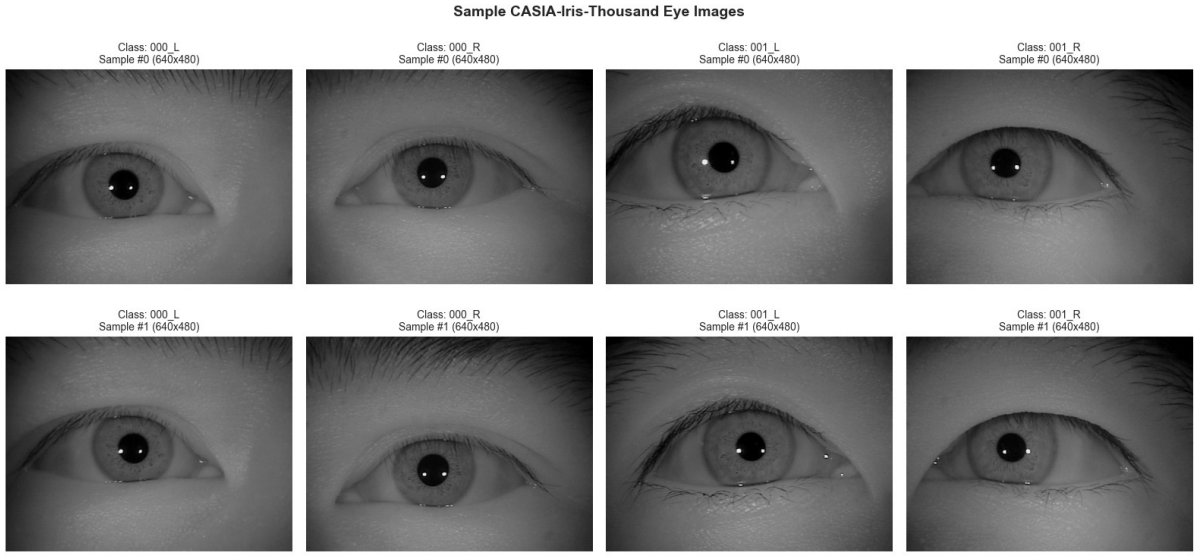


Figure 2. Sample near-infrared iris images from the dataset

The images were also analyzed for quality using measures such as brightness, contrast, sharpness, reflections and darkness, as shown in Figure 3. These checks helped assess the dataset quality but were not used for training the models.

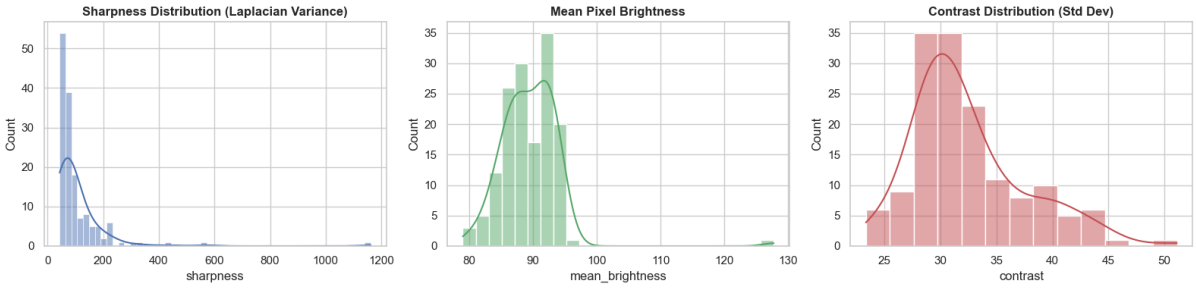


Figure 3. Sampled image quality metric profiles

4.3 Iris Preprocessing and Normalization

The preprocessing stage includes iris detection, normalization and image enhancement. The iris region is converted into a standard 64 × 256 grayscale strip using polar normalization, and

contrast-limited adaptive histogram equalization (CLAHE) is applied to improve image quality, as shown in Figure 4. During preprocessing, all 20,000 images were processed successfully, with no failures or skipped images. The final images were saved as 8-bit grayscale PNG files along with metadata linking each processed image to its original identity and image path.

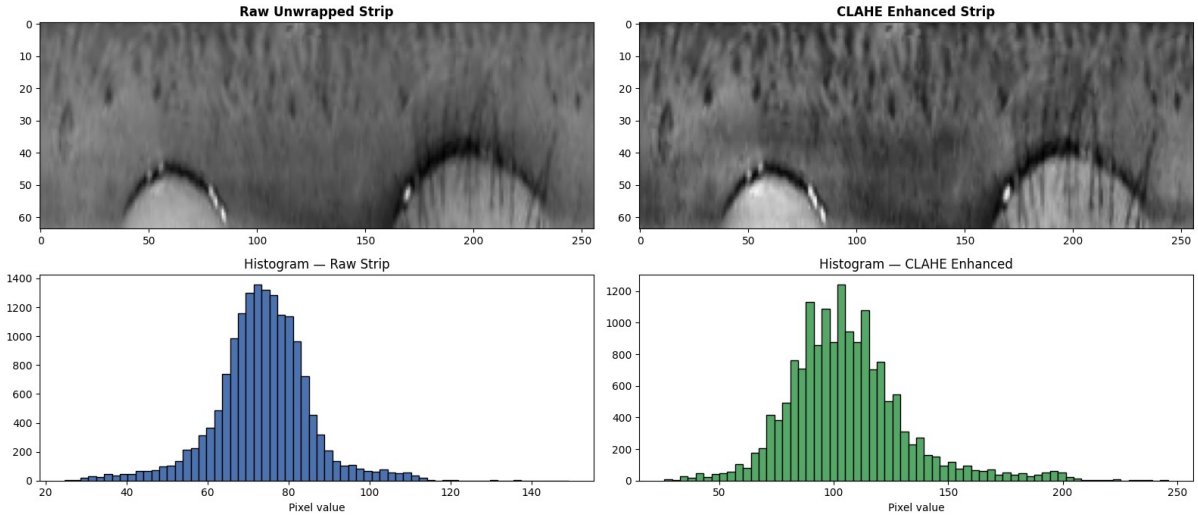


Figure 4. Iris segmentation, normalization and enhancement stages

4.4 Identity-Disjoint Split and Few-Shot Protocol

The dataset was divided into 1,400 training identities, 300 validation identities and 300 test identities, resulting in 14,000, 3,000 and 3,000 images respectively, as shown in Table 2 and Figure 5. Identity overlap between the three splits was checked and removed to ensure a fair evaluation. For few-shot learning, 1-shot, 3-shot and 5-shot enrollment protocols were created. In the test set, the support set contains 300, 900 and 1,500 images respectively, while the remaining images are used as query samples for evaluation.

Table 2. Identity-disjoint split used for the metric-learning experiments

Split	Identities	Images	Images per identity
Training	1,400	14,000	10
Validation	300	3,000	10
Test	300	3,000	10
Total	2,000	20,000	10

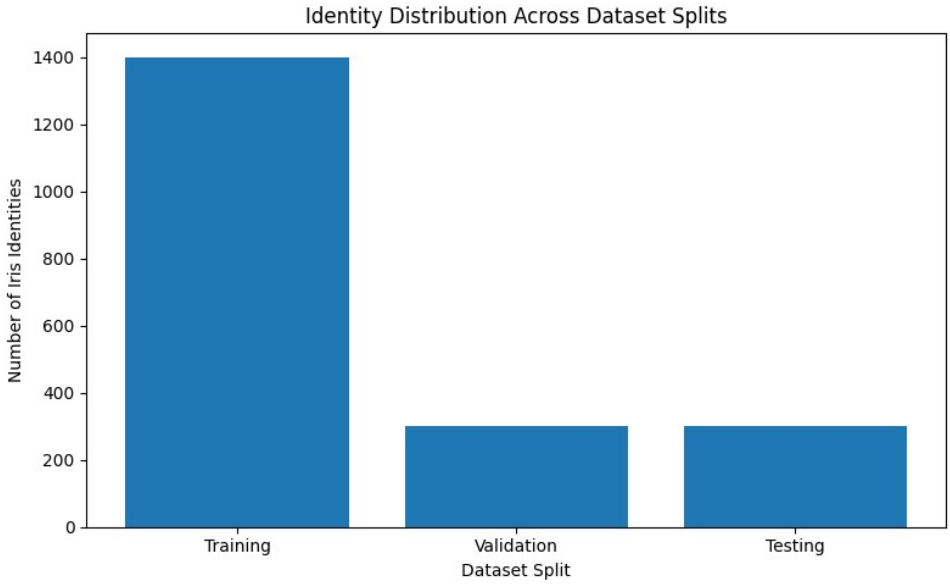


Figure 5. Identity-disjoint training, validation and test split

4.5 Classical CNN Baseline

A ResNet34-based baseline model was trained for closed-set iris classification using 1,400 identities. The model uses a 512-dimensional feature representation along with an ArcFace-style classification head. The best validation accuracy achieved was 67.0%, while the test results showed 71.7% Rank-1 accuracy, 72.1% Rank-1 accuracy with test-time augmentation and 77.8% Rank-5 accuracy. The training and validation loss curves are shown in Figure 6.

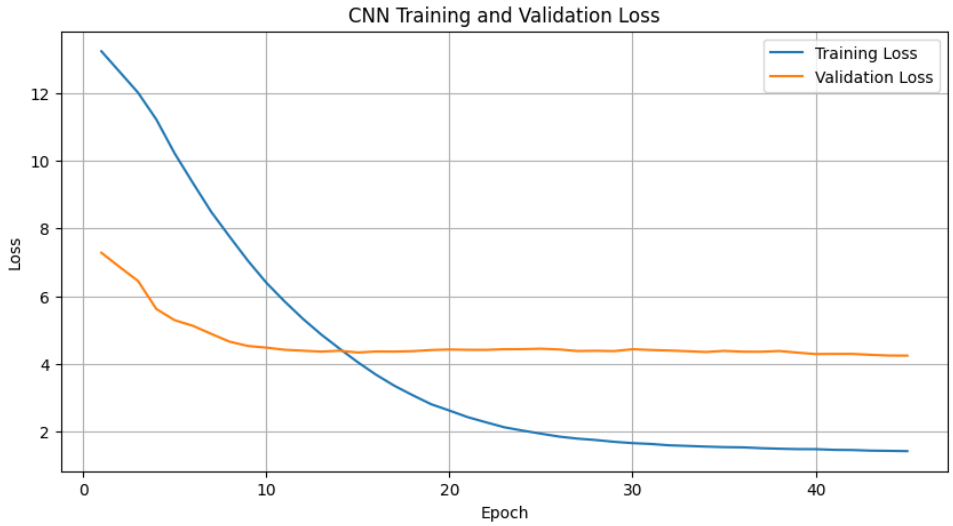


Figure 6. Classical ResNet34 training and validation curves

These results serve as a baseline for comparison. Since the same identities are used in both training and testing in this setup, the results come from a closed-set classification task and should not be compared directly with the later identity-disjoint verification experiments, in which the test identities are completely unseen during training.

4.6 Siamese Metric Learning

A Siamese network based on ResNet34 was trained to learn a 128-dimensional iris embedding from pairs of images. The model was evaluated using identity-disjoint data, ensuring that the test identities were not seen during training. The best model was obtained at epoch 60, achieving a validation AUC of 0.8831 and a validation EER of 20.78%, as shown in Figure 7. On the test set, it achieved an accuracy of 79.28%, an AUC of 0.8797 and an EER of 21.92%. These results serve as the main 128-dimensional metric learning baseline for comparison with the compact 8-dimensional and hybrid quantum-classical models.

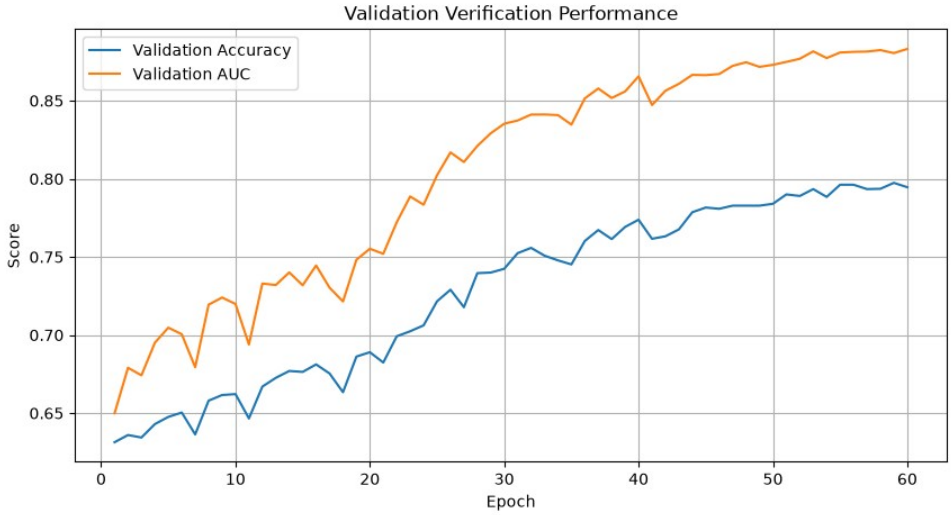


Figure 7. Siamese metric-learning training curves

4.7 Classical Projection Baseline

Notebook 6 reduces the 128-dimensional embedding to a compact 8-dimensional embedding using a small projection network. The model was trained and evaluated on the same validation and test pairs used for the Siamese baseline, allowing a fair comparison.

The best model was obtained at epoch 16 and achieved 78.58% accuracy, an AUC of 0.8747 and an EER of 22.18% on the test set (Figure 8). The results show that the embedding size was greatly reduced with only a small decrease in performance compared with the original 128-dimensional Siamese model.

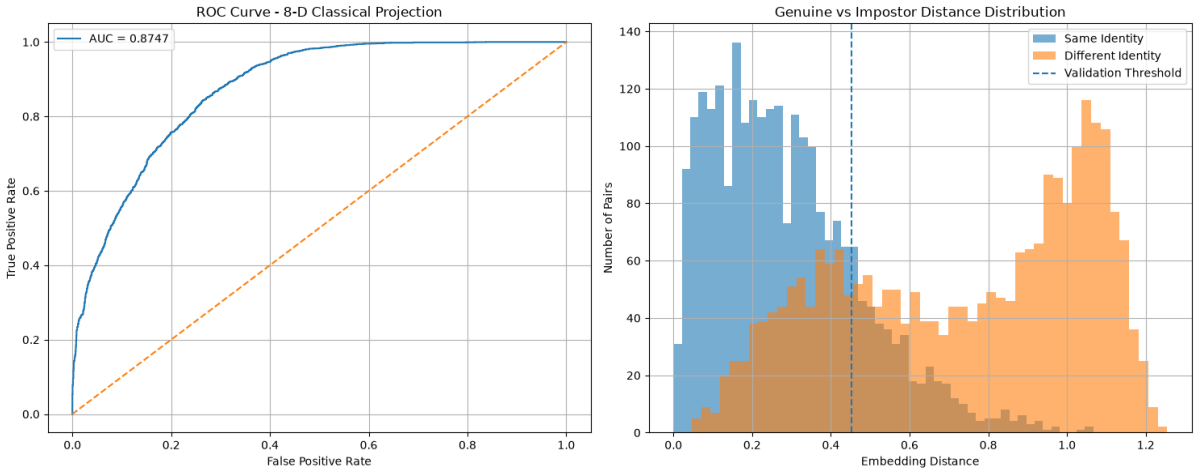


Figure 8. Classical 128-D to 8-D projection results

4.8 Quantum Circuit Experiments

Notebook 7 uses the 8-dimensional embeddings generated by the classical projection model as input to an 8-qubit quantum circuit. Different circuit designs were tested by changing factors such as circuit depth and entanglement patterns, as shown in Figure 9. The quantum circuit was compared with a classical linear model and a classical multilayer perceptron (MLP) using the same input features, and the results showed very similar performance across all methods (Figure 10). The classical MLP achieved the best accuracy (79.12%) and AUC (0.8769), while the quantum model achieved 78.68% accuracy and an AUC of 0.8755. These results indicate that the quantum model did not outperform the best classical model in the frozen-feature experiment.

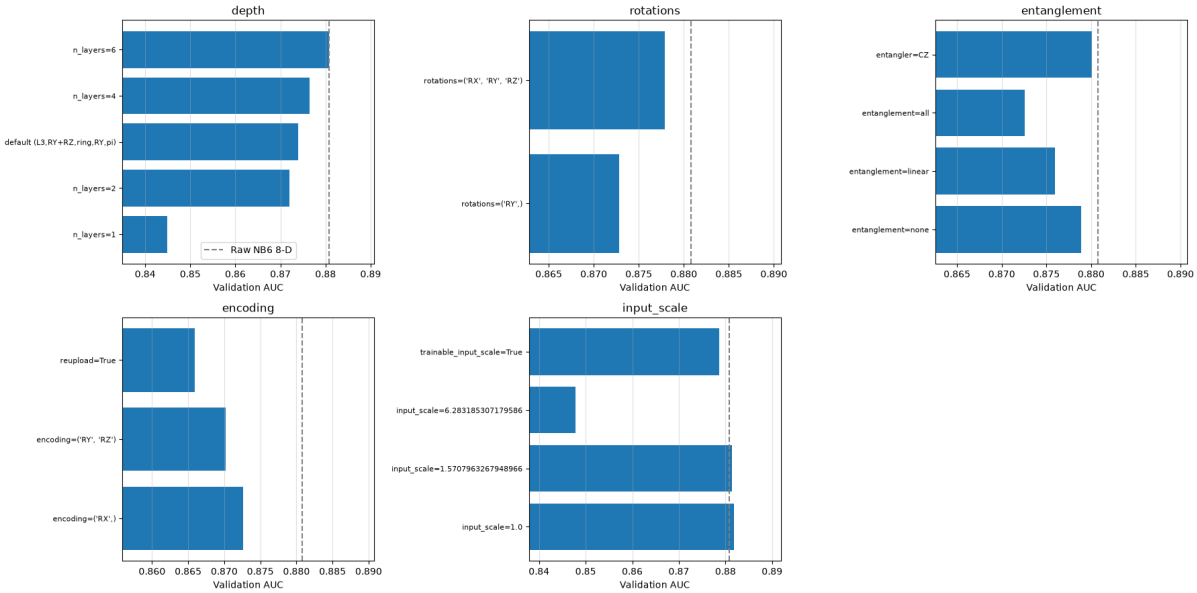


Figure 9. Quantum circuit design sweep results

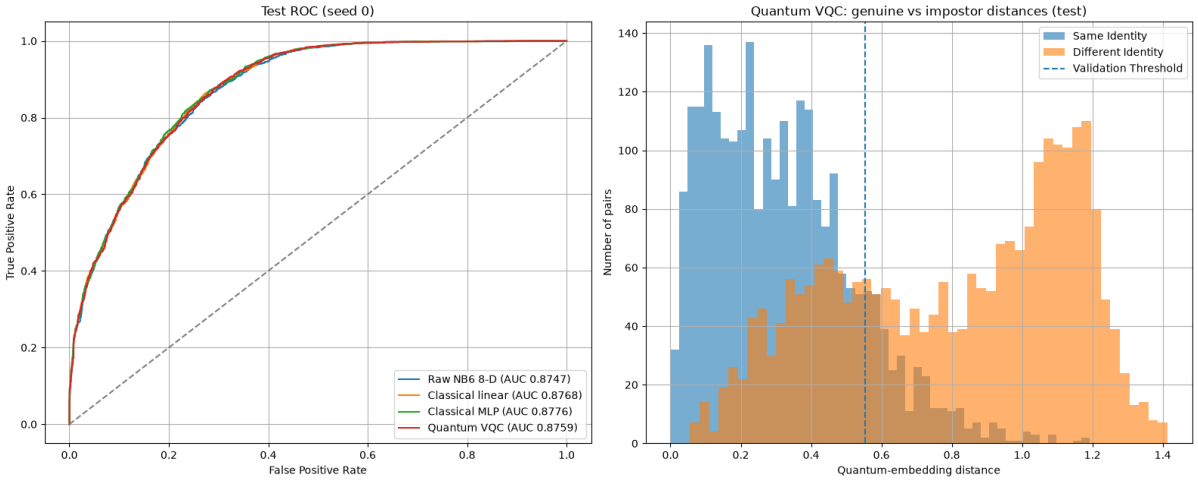


Figure 10. Frozen-feature classical and quantum head comparison

4.9 End-to-End Hybrid Metric Learning

Notebook 8 trains the encoder and metric head together using three variants: classical linear, hybrid quantum and hybrid quantum product. All models were evaluated on the same identity-disjoint validation and test sets. The hybrid quantum model achieved the best accuracy and AUC, with 79.90% accuracy, an AUC of 0.8813 and an EER of 21.38%. The classical linear model achieved 78.72% accuracy, an AUC of 0.8761 and an EER of 21.58%, while the hybrid quantum product model achieved 79.42% accuracy, an AUC of 0.8803 and an EER of 20.98%. The training curves are shown in Figure 11 and the test ROC curves are shown in Figure 12. These results suggest that the hybrid quantum models are competitive and slightly outperform the classical model in this experiment. However, the differences are small and should be interpreted together with the later bootstrap and ablation analysis, rather than as proof that quantum models are always better.

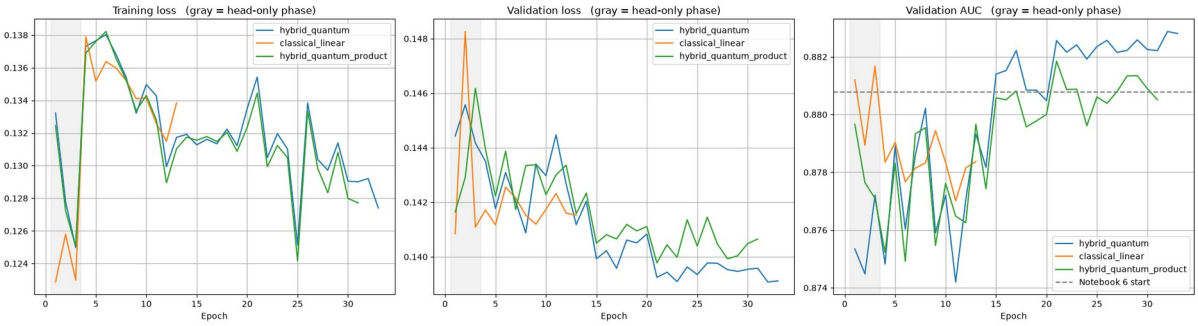


Figure 11. End-to-end hybrid model training curves

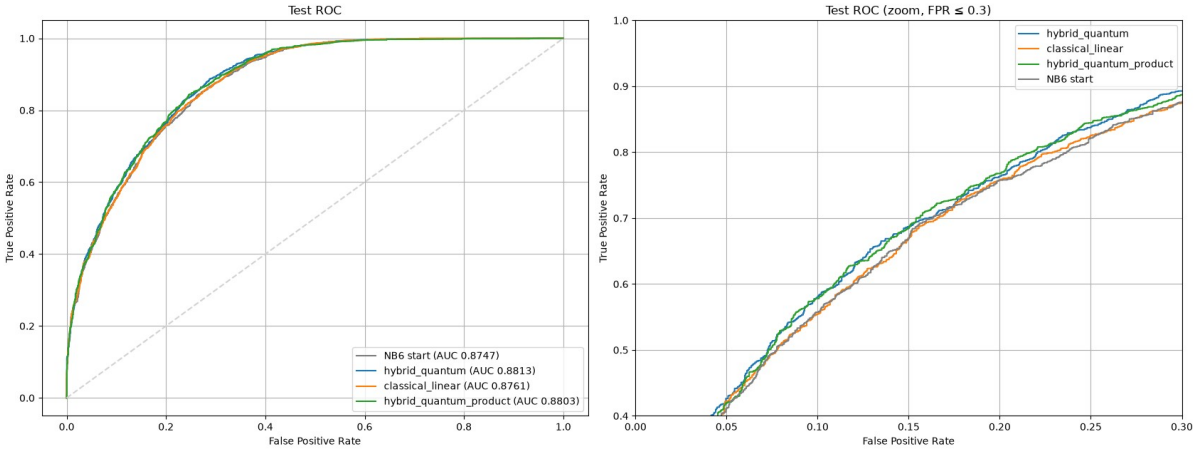


Figure 12. Hybrid model test ROC curves (full and zoomed views)

4.10 Few-Shot Evaluation and Final Ablation

Notebook 9 evaluates the models using 200 few-shot test episodes for the 1-shot, 3-shot and 5-shot settings. It reports metrics such as Rank-1, Rank-5, AUC and EER, as shown in Figure 13. Notebook 10 performs a more detailed evaluation using a larger set of test pairs and 1,000 bootstrap samples for statistical analysis. Since it uses a different evaluation protocol, its results are slightly different from those reported earlier. Therefore, the results from each notebook should be reported separately and compared only within their respective evaluation settings.

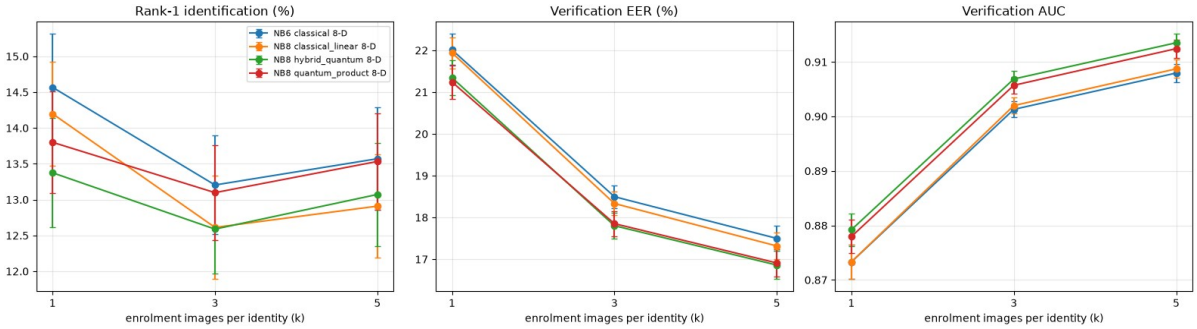


Figure 13. Few-shot evaluation curves for 1-, 3- and 5-shot protocols

5. Proposed Algorithms

5.1 Algorithm A: Identity-Disjoint Dataset Preparation

1. Extract image information such as the subject ID, eye identity and image path.
2. Check the image quality, readability, duplicates and dataset statistics.
3. Preprocess all iris images into a standard 64 × 256 format.
4. Split the data by identity into training, validation and test sets.
5. Assign 1,400 identities for training, 300 for validation and 300 for testing.
6. Verify that there is no identity or image overlap between the splits.
7. Create and validate 1-shot, 3-shot and 5-shot support and query sets for few-shot learning.

5.2 Algorithm B: Siamese Metric-Learning Verification

8. Load pairs of preprocessed iris images and their labels.
9. Pass both images through a shared ResNet34 encoder to generate 128-D embeddings.
10. Compare the embeddings and calculate the metric learning loss.
11. Train the model and monitor the validation loss, AUC and EER.
12. Select the best model and threshold using the validation set.
13. Evaluate the final model on the test set and report accuracy, AUC, EER, FAR and FRR.

5.3 Algorithm C: Hybrid Quantum Metric Head

14. Generate an 8-D embedding from the processed iris image.
15. Encode the embedding into an 8-qubit quantum circuit.
16. Apply trainable quantum operations and entanglement between the qubits.
17. Measure the circuit to produce a new 8-D feature vector.
18. Train the model using genuine and impostor image pairs.
19. Update the encoder, projection layer and quantum circuit parameters during training.
20. Select the best model using the validation results and evaluate it on the test set.

5.4 Algorithm D: Episodic Few-Shot Identification

21. Select 1-shot, 3-shot or 5-shot enrollment and choose the support images for each identity.
22. Generate embeddings and create an identity template from the support images.
23. Extract embeddings for the query images and compare them with the enrolled templates.

24. Rank identities based on similarity scores and calculate the Rank-1 and Rank-5 accuracy.
25. Evaluate verification performance using AUC, EER and the true acceptance rate (TAR).
26. Repeat the process across multiple test episodes and report the average results.

5.5 Algorithm E: Identity-Level Paired Bootstrap

27. Prepare common comparison pairs for all evaluated models.
28. Test each model using thresholds selected from the validation data.
29. Apply bootstrap sampling on the test identities to estimate the variation in performance.
30. Evaluate all models on the same set of sampled identities.
31. Calculate AUC, EER and 95% confidence intervals for each model.
32. Compare the results and report differences only when they are statistically meaningful.

6. Proposed Flowchart / Block Diagram

6.1 System Block Diagram

The system follows a step-by-step process that includes data preparation, feature learning, model training and performance evaluation, as shown in Figure 14. The workflow used in this project is based on the submitted notebooks, and the block diagram provides a simple overview of the complete process.

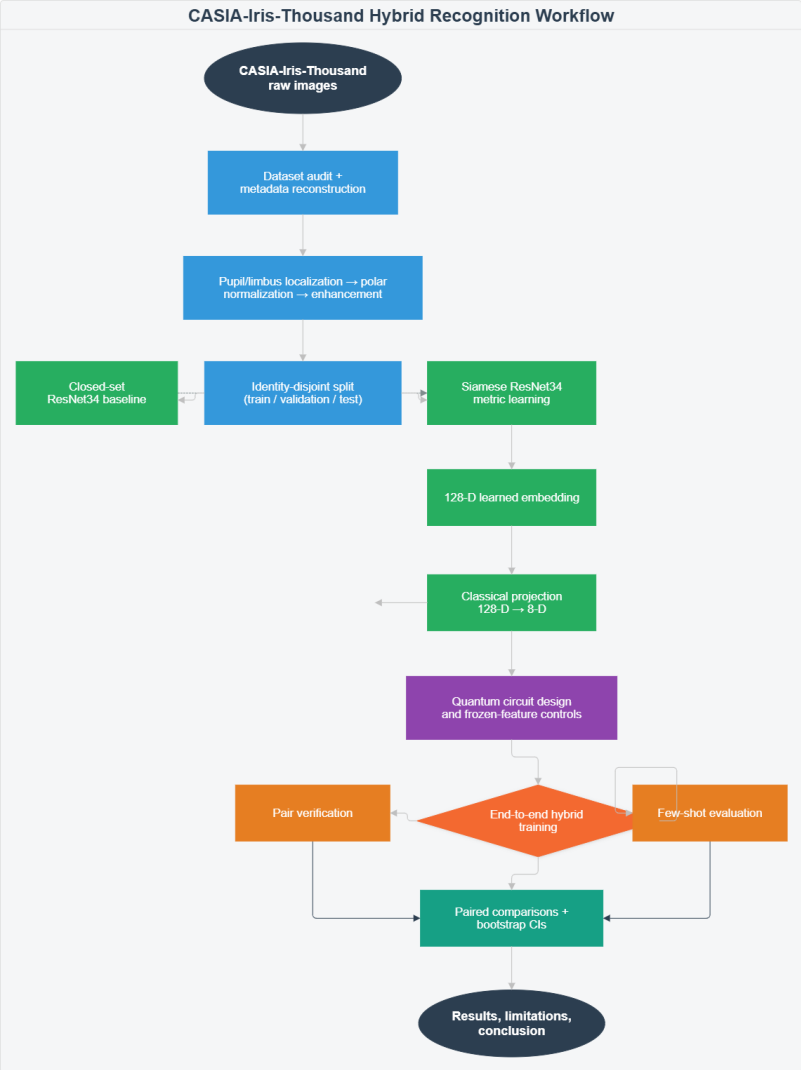


Figure 14. System block diagram

6.2 Data Flow Description

The system takes raw iris images and identity information as input. The images are first preprocessed and converted into normalized iris strips. Next, the data is divided into training, validation and test sets with no identity overlap. The training data is used to train the models, the validation data is used for model and threshold selection, and the test data is used for the final assessment. The hybrid models use a quantum circuit simulator instead of real quantum hardware. Finally, the system assesses performance through verification scores, identification accuracy and statistical analysis.

6.3 Measures to Prevent Information Leakage

- Divide the dataset by identity into separate training, validation and test sets.
- Choose the optimal threshold using the validation set only.
- Compare all models on the same test pairs for a fair comparison.
- Separate the support images and query images in the few-shot experiments.
- Report confidence intervals, and do not use test results to guide training or model selection.

7. Tools, Technologies and Implementation Details

7.1 Software Stack

Table 3 lists the main methods and software used at each stage of the project, as evidenced in the notebooks. The models were optimized using AdamW, a variant of the Adam optimizer [25], together with learning-rate scheduling and validation checkpoints.

Table 3. Research workflow

Stage	Main methods and software evidenced in the notebooks	Purpose
Exploration	Python, pandas, NumPy, Matplotlib, image I/O	Audit counts, quality and image structure
Preprocessing	OpenCV/image-processing routines, CLAHE, polar normalization	Produce normalized iris strips
Deep learning	PyTorch, Torchvision, ResNet34, CUDA	CNN and Siamese representation learning
Optimization	AdamW, learning-rate scheduling, validation checkpoints	Train and select model variants
Quantum simulation	Custom differentiable simulator; RY/RZ/CNOT operations	Build an 8-qubit variational metric head
Evaluation	ROC/AUC, EER, FAR/FRR, Rank-1/Rank-5, bootstrap	Measure verification and identification
Persistence	CSV, JSON configuration, model checkpoints, NumPy embeddings	Reproducibility and transfer between notebooks

7.2 Data and Metadata Architecture

The project uses metadata tables containing the subject ID, eye side, identity label, image name and image path. The data is partitioned into 14,000 training, 3,000 validation and 3,000 test images. Data is passed between the different parts of the project in the form of 8-D and 128-D embeddings, pair lists, model checkpoints and configuration files. These files should be stored and named appropriately so that the same files can be generated again for reproducibility. To port the project from one machine to another, absolute paths should be avoided in the code and only relative paths should be used.

7.3 Iris Image Preprocessing Framework

In the preprocessing step, the iris area is identified, reflection areas are removed and the iris is transformed into a 64 × 256 pixel grayscale image, which is then enhanced using CLAHE. All images were stored as grayscale PNG images for the CNN. From a sample batch, the processing time was around 2.4 images per second, which means that the complete set can be processed in an acceptable amount of time.

7.4 CNN-Based Metric Learning

The CNN baseline uses ResNet34 for iris classification, and the Siamese model learns a 128-dimensional embedding for verification. A projection network then reduces the embedding to 8 dimensions. The best models are selected using the validation results, and the same test pairs are reused whenever possible for a fair comparison.

7.5 Quantum Circuit Implementation

The selected quantum circuit uses 8 qubits and 3 trainable layers. The input features are encoded into the circuit, processed through trainable quantum operations and measured to produce an 8-dimensional output. The circuit contains 48 trainable parameters. Several tests were performed to verify that the simulator was working correctly. Since all experiments were run on a quantum simulator, these results validate the implementation but do not prove any real quantum advantage.

7.6 Evaluation Metrics

Table 4 describes the evaluation metrics used in this project and how each of them should be interpreted.

Table 4. Evaluation metrics and interpretation

Metric	Interpretation	Preferred direction
Accuracy	Fraction of pair decisions correct at a selected threshold	Higher
AUC	Pair-ranking quality across thresholds	Higher
EER	Error point where FAR and FRR are equal or closest	Lower
FAR	Fraction of impostor comparisons accepted as genuine	Lower
FRR	Fraction of genuine comparisons rejected	Lower
TAR@FAR	Genuine acceptance rate at a specified FAR	Higher
Rank-1	Correct identity is the highest-ranked template	Higher
Rank-5	Correct identity appears among the top five templates	Higher

7.7 Reproducibility and Configuration

The experiments were conducted using a fixed random seed of 42 to ensure reproducibility. Model configurations, embeddings, checkpoints and evaluation results were saved for later analysis.

For evaluation, the project used 200 few-shot test episodes for the 1-shot, 3-shot and 5-shot settings, while the final ablation study used 100 additional episodes and 1,000 bootstrap

samples to estimate uncertainty. The experiments were run using PyTorch with CUDA support on an NVIDIA GeForce GTX 1650 GPU.

8. Results, Discussion and Comparative Analysis

8.1 Exploratory Data Analysis

The dataset audit reported that the dataset consisted of 20,000 images from 1,000 subjects and 2,000 eye identities, with 10 images per identity and an equal number of right-eye and left-eye images. No duplicate or unreadable images were encountered during the quality check. At the preprocessing stage, all 20,000 images were processed successfully, and sample checks verified that the output images had the planned shape. These checks provided confidence that the data being prepared was consistent.

8.2 Classical CNN Baseline

The closed-set ResNet34 classifier achieved a test Rank-1 accuracy of 0.7171 and a Rank-5 accuracy of 0.7779, and test-time augmentation raised the Rank-1 accuracy to 0.7214. A validation accuracy of 0.6700 was achieved at epoch 30. These results are for the internal closed-set split, in which the training, validation and closed-set test images all include the same 1,400 identities. They should be treated as a classification baseline and not as an approximation of unseen-identity verification.

8.3 Metric Learning and Comparison Results

Table 5 summarizes the key experimental findings from the notebooks.

Table 5. Key experimental findings

Model / notebook protocol	Embedding dim.	Accuracy	AUC	EER
Notebook 5: Siamese metric learning	128	0.7928	0.8797	21.92%
Notebook 6: classical projection	8	0.7858	0.8747	22.18%
Notebook 7: frozen-feature classical linear head	8	0.7899	0.8765	21.67%
Notebook 7: frozen-feature classical MLP head	8	0.7912	0.8769	21.68%
Notebook 7: frozen-feature quantum VQC head	8	0.7868	0.8755	21.81%
Notebook 8: end-to-end hybrid quantum	8	0.7990	0.8813	21.38%
Notebook 8: end-to-end classical linear	8	0.7872	0.8761	21.58%
Notebook 8: end-to-end quantum product	8	0.7942	0.8803	20.98%

The entries in Table 5 are reproduced from their respective notebook summaries and should not be considered perfectly comparable, as the training procedures differ between the stages and some later analyses use different pair sets. Between Notebook 5 and Notebook 6, the 8-D projection forces a reduction from 128 to 8 dimensions, with the AUC falling from 0.8797 to 0.8747 and the accuracy falling from 0.7928 to 0.7858. The difference is modest, but it is consistent with a compactness/performance trade-off.

8.4 Frozen-Feature Quantum Comparison

Table 6 compares the quantum circuit with the classical frozen-feature heads, and Figure 15 shows the corresponding ROC curves and distance distributions.

Table 6. Quantum circuit and frozen-feature control comparison

Head	Trainable head parameters	Mean test AUC	Mean test accuracy
Classical linear	64	0.8765	0.7899
Classical MLP	280	0.8769	0.7912
Quantum VQC	48	0.8755	0.7868

The frozen-feature experiment showed that the variational quantum circuit (VQC) model did not outperform the best classical models in terms of test accuracy or AUC. Although the quantum model used fewer trainable parameters, having fewer parameters alone does not mean that it is more efficient or better. Since no training time, runtime or energy consumption analysis was performed, the results do not provide evidence that the simulated quantum model is computationally cheaper than the classical alternatives.

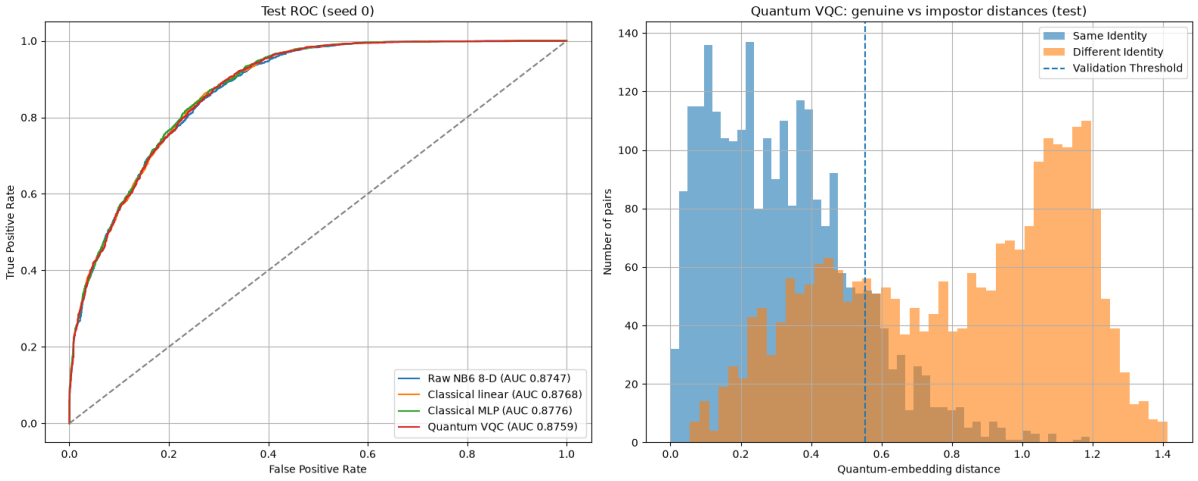


Figure 15. Frozen-feature classical and quantum head comparison

8.5 End-to-End Hybrid Model Results

In Notebook 8, the hybrid quantum model achieved the best AUC (0.8813) and accuracy (79.90%), while the hybrid quantum product model achieved the lowest EER (20.98%). This

shows that model superiority depends on the metric considered. The training curves and test ROC curves are shown in Figures 16 and 17 respectively.

The bootstrap analysis showed that the hybrid quantum model obtained a small but statistically significant improvement in AUC over the classical model in this evaluation setup, whereas the quantum model did not outperform the best classical model in the earlier frozen-feature experiment (Notebook 7). We hypothesize that the improvement seen in Notebook 8 is a result of the combined training procedure, which optimizes both the feature extractor and the metric head jointly, rather than of the quantum circuit on its own.

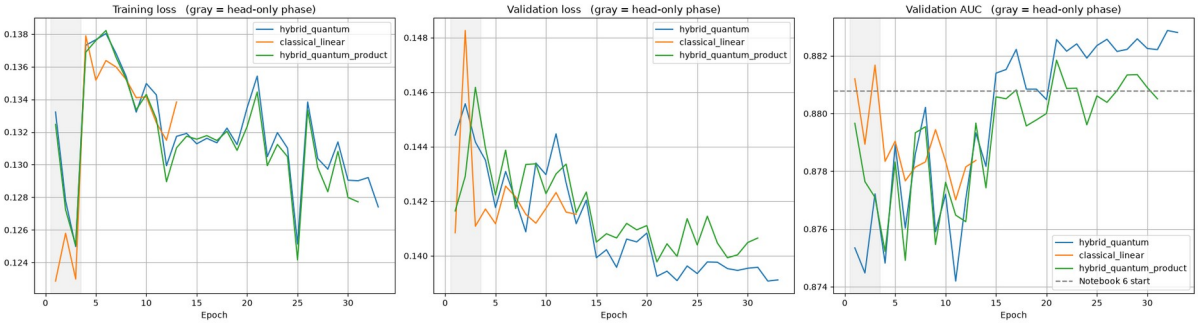


Figure 16. End-to-end hybrid model training curves

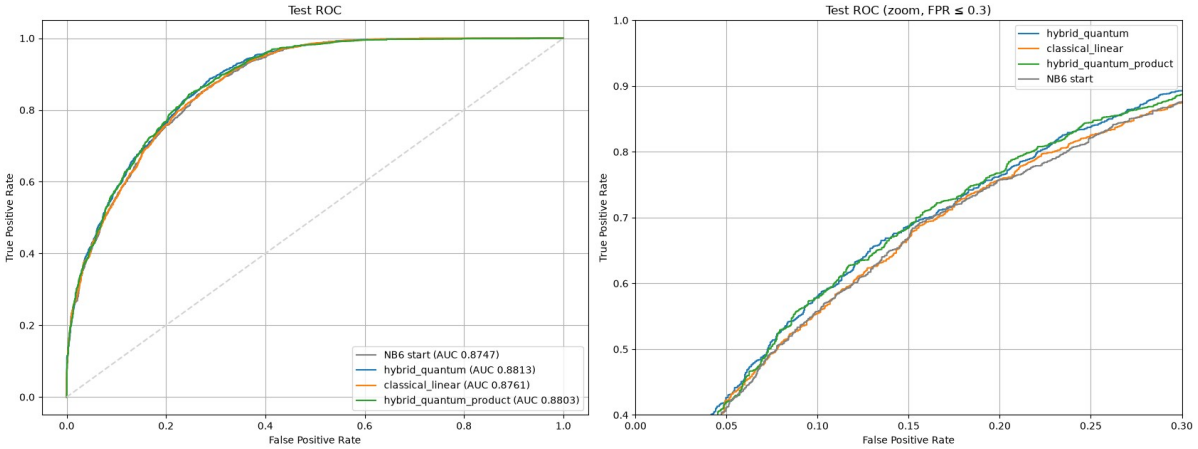


Figure 17. Hybrid model test ROC curves (full and zoomed views)

8.6 Final Verification Results and Confidence Intervals

Notebook 10 performed the final comparison using shared test pairs and bootstrap analysis, as reported in Table 7 (NB denotes Notebook). The 128-D Siamese model achieved an AUC of 0.8797 and an EER of 21.36%, while the 8-D hybrid quantum model achieved an AUC of 0.8791 and an EER of 21.35%. The quantum product model achieved the lowest EER of 21.21%.

Table 7. Final paired-pair verification results

Model	Dim.	AUC	EER	FAR at validation threshold	FRR at validation threshold
NB6 encoder	128	0.8797	21.36%	21.65%	21.04%
NB6 classical projection	8	0.8732	22.02%	22.12%	21.87%
NB8 classical linear	8	0.8731	21.93%	21.94%	21.90%
NB8 hybrid quantum	8	0.8791	21.35%	21.98%	20.60%
NB8 quantum product	8	0.8778	21.21%	22.11%	20.02%

The bootstrap confidence intervals, shown in Figures 18 and 19, showed considerable overlap, indicating that the performance differences between the models are small. Overall, the hybrid quantum models performed competitively with the 128-D baseline while using a compact 8-dimensional representation.

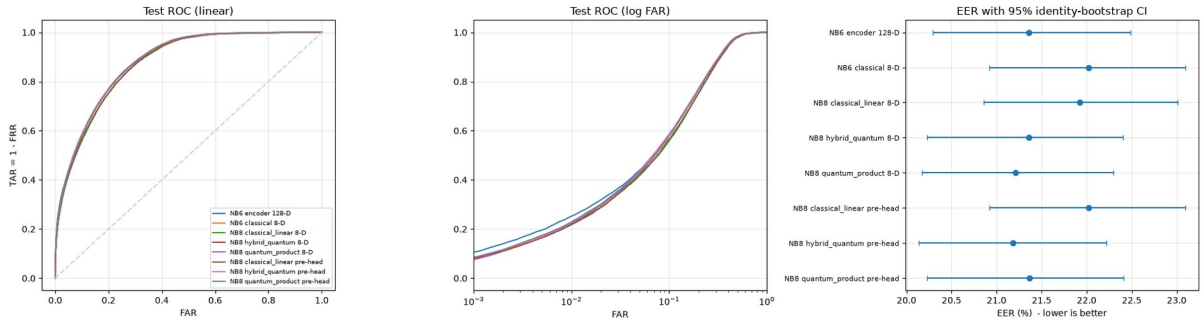


Figure 18. Verification ROC curves and EER confidence intervals

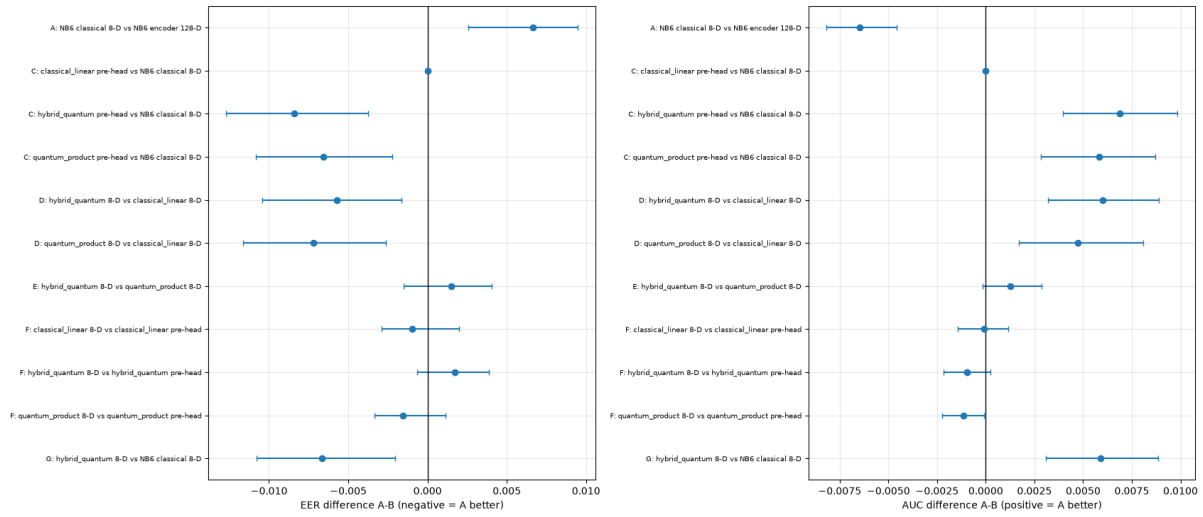


Figure 19. Paired ablation effects with confidence intervals

8.7 Few-Shot Evaluation

Notebook 9 evaluates 200 episodes per shot count and compares four 8-D models. In the 1-shot setting, the reported Rank-1 rates are 14.57% for the Notebook 6 classical model, 14.20% for the end-to-end classical linear model, 13.37% for the hybrid quantum model and 13.80% for the quantum product model. The corresponding AUC values are 0.8733, 0.8733, 0.8792 and 0.8780, while the EER values are 22.01%, 21.94%, 21.34% and 21.24% (Figure 20). This illustrates that verification ranking metrics and multi-class identification metrics need not select the same model. The paired episode comparisons in the notebook show that the hybrid quantum model has a lower EER than the classical linear model in the 1-shot and 3-shot comparisons, but the Rank-1 differences are not well separated in those same comparisons.

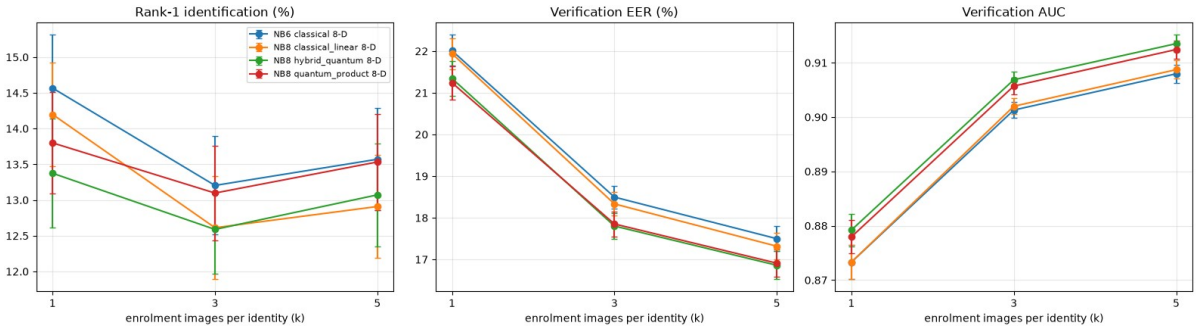


Figure 20. Few-shot evaluation curves for 1-, 3- and 5-shot protocols

8.8 Interpretation of Ablations

The final ablation study showed that reducing the embedding size from 128 to 8 dimensions led to a small decrease in performance. In the 8-D classical model, the EER was roughly 0.66 percentage points higher than that of the 128-D model, which shows a trade-off between compactness and accuracy (Table 7 and Figure 19). It was also observed that inserting the quantum circuit without joint training does not necessarily improve the model, so joint training and model design are also important. The Rank-1 performance by enrollment shots and the FAR/FRR trade-off are shown in Figure 21. Therefore, the results should be interpreted carefully and supported by the reported confidence intervals.

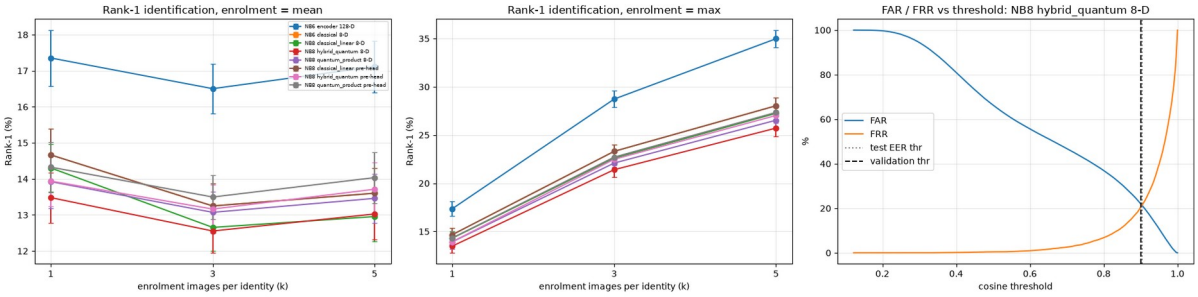


Figure 21. Rank-1 performance by enrollment shots and FAR/FRR trade-off

8.9 Threats to Validity and Limitations

Table 8 summarizes the threats to validity and the limitations of this work, together with the mitigation or reporting action for each.

Table 8. Limitations and risk mitigation

Limitation	Implication	Mitigation / reporting action
Different pair protocols across notebooks	Point estimates may differ even for the same model family	Label each metric with its notebook and protocol; prioritize the shared-pair Notebook 10 for the final comparison
Single seed for end-to-end hybrid variants	Training variance is not fully characterized	Repeat end-to-end training across multiple seeds
Classical simulation of the quantum circuit	No evidence of real quantum hardware advantage	Describe as a simulated variational circuit; test on hardware only as future work
Only the CASIA-Iris-Thousand dataset	Generalization to other sensors and datasets is unknown	Evaluate on additional public iris datasets and sensor conditions
Modest few-shot Rank-1 values	Identification remains challenging with few support samples	Report uncertainty and explore enrollment aggregation, augmentation and calibration
Potential implementation and data-path issues in the early split notebook	Raw filename parsing warnings occurred before metadata reconstruction	Document the reconstructed metadata stage and re-run end-to-end checks from a clean environment
Threshold dependence	FAR, FRR and accuracy depend on the operating threshold	Select the threshold on validation data and report fixed-threshold test metrics

9. Conclusion and Future Scope

9.1 Conclusion

This project developed and evaluated an iris recognition system using the CASIA-Iris-Thousand dataset. All 20,000 iris images were successfully preprocessed and divided into identity-disjoint training, validation and test sets.

The 128-dimensional Siamese model achieved strong verification performance, while the 8-dimensional projection model reduced the feature size with only a small drop in accuracy. Quantum-based models were also evaluated. The frozen-feature quantum model did not outperform the best classical model, but the end-to-end hybrid quantum model achieved competitive results and improved some evaluation metrics.

These results indicate that hybrid quantum-classical models could be a feasible approach, but they do not prove that quantum approaches will always outperform classical approaches. Hence, this project provides a careful comparison of classical and quantum approaches using compact embeddings, few-shot learning and statistical analysis.

9.2 Future Scope

- Train the models on several random seeds and report the average results.
- Test the models on other iris datasets to see how they perform.
- Run the quantum model on real quantum hardware and compare the results with the simulator results.
- Try different quantum circuit designs and layouts.
- Compare other compact embedding methods to find the better one.
- Improve few-shot learning through more accurate enrollment and matching.
- Assess model robustness under changing lighting conditions, with different sensors and at varying image quality levels.
- Focus on the whole pipeline with clear documentation and a reproducible setup so that it is easy to deploy and test.

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11. Plagiarism Report